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Public spending, public governance and economic growth at the Vietnamese provincial level: A disaggregate analysis

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ABSTRACT

This study examines the role of economic governance in the relationship between public spending, private investment, and economic growth in Vietnam at the provincial level. The study data consist of sixty-two Vietnamese provinces for the period 2006–2015. Some notable results are attained by applying a sequential (two-stage) estimation. First, the marginal benefits to economic growth of increased Vietnamese provincial government expenditures may be constrained because of the inefficiency of expenditures on education, business services, and public administration. Second, public spending and private investment are found to be substitutes at the provincial level. Third, based on the Provincial Competitiveness Index (PCI) ranking, used as a proxy for provincial public governance in Vietnam, the significance of public governance can be observed. The study concludes that good governance, characterized by different attributes, such as lower informal charges, greater transparency, and unbiased policy, plays a critical role in improving the impact of government expenditure on economic growth in Vietnamese provinces, particularly through its interactions with private sector investment.

1. Introduction

Private investment makes an important contribution to social and economic development. Many studies have focused on private investment in terms of economic growth (Mallick, 2013; Menon, 2014; Morrissey and Udomkermongkol, 2012). From the perspective of endogenous growth theory, governments are concerned with how private investment reacts to movements in government policy in order to implement economic development programs. As argued by Blejer and Khan (1984), an increase in private investment has a positive effect on economic growth, all else unchanged. However, this result is generally ambiguous when governments influence private investment decisions via changes in their policies. It is argued that the linkage between public spending and outcomes could become serious because increased public provision could crowd out private sector provision. This raises the question of the efficiency of public spending per se in terms of budget constraints. What governments should do instead is to substitute public spending for private investment now that additional public spending could reduce the marginal effect on outcomes (Curristine et al., 2007; Rajkumar and Swaroop, 2008). It is argued that the quality of governance is critical for public spending and outcomes to be efficient. The inefficacy of public spending can be attributed to institutional inefficiency, which explains why governments with weak institutional capacity face difficulty in transforming public spending into effective public goods provision (Rajkumar and Swaroop,

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2008). Several economic models have captured aspects of good governance such as controlled corruption, transparency, and greater accountability, which can affect public spending and outcomes (Dzhumashev, 2014; Kumah and Brazys, 2016; Rajkumar and Swaroop, 2008). However, the role of governance in shaping the linkage between public spending and outcomes is still a matter of debate in the empirical literature (Hu and Mendoza, 2013).

Further, several writers have focused on the role of public governance in the connection between public spending and outcomes at the national or subnational level (Di Liddo et al., 2018; Mejia Acosta and Tillin, 2019; Nguyen et al., 2019; Shon and Cho, 2019; Wilson, 2016; Zhang and Zou, 1998). Vietnam, as an emerging economy, thus provides an ideal framework for empirically examining the linkages among public governance, public spending, private investment, and economic growth. This paper builds on previous studies on similar topics in the following ways. First, its primary purpose is to address the relationship between public spending, private investment, and economic growth at the provincial level. The model in this study is driven by the production function of a province in which local public spending and public governance are endogenous, employed as components of total factor productivity (TFP). These endogenous factors are assumed to affect input and output growth, instead of the remaining parts of TFP. Public spending produces basic public services/goods, which significantly support private producers. By contrast, public spending can limit the productive potential of private investment if local authorities use unproductive public resources. From the perspective of institutional theory, public governance is considered the demonstration of government power in managing social and economic resources in terms of good governance, such as reduced informal charges, transparency and greater accountability, and unbiased policy, which are fundamental to reducing transaction costs, asymmetric information, and risks and then stimulating private sector investment.

Second, the study aims to address how economic governance, measured by the level of informal charges, transparency, and policy bias, affects the growth effects of public spending and private investment. Nguyen et al. (2019) showed that better governance and public administration condition improvements in income allocation and poverty reduction in Vietnam. In addition, local institutions such as controlled corruption significantly affect the business strategy and returns of local private enterprises (Nguyen and Van Dijk, 2012; Nguyen et al., 2013). Existing studies have largely overlooked the role that public governance plays in influencing private investment and the nature of the association between government spending and private investment, which are fundamental to economic growth. Some criticism has been expressed about these developments, with a widely held perception that fiscal decentralization and local public spending lack efficiency (Nguyen and Anwar, 2011). In particular, Tran et al. (2009) indicated that transparency and accountability have lagged behind the rapid pace of decentralization. Effective linkages between institutions and the private sector are often difficult to establish because of the fragmented nature of the market participants, whereas other issues, such as uneven legal implementation, policy bias, and lack of transparency, have been formidable barriers to private sector investment. Therefore, we put forward the central proposition that managing public spending to enhance the growth effect of private investment requires local authorities to enhance the quality of governance in line with the reduction of informal charges, transparency and greater accountability, and unbiased policy. The empirical model developed in the paper is intended to offer some insights into the significance of this proposition. Better growth performance, with greater private sector involvement, opens up further opportunities for investigating ways in which the quality of governance can be enhanced.

The study uses a set of panel data on sixty-two Vietnamese provinces over the period 2006–2015. Economic growth in Vietnam slowed down during this period because of the 2018 financial crisis. Therefore, the Vietnamese government implemented a comprehensive plan on economic restructuring. For example, the government significantly reduced the size of public investment and increased public spending on education, health, the environment, and scientific activity. At the same time, fiscal decentralization has been largely devolved to local authorities based on accountability and efficiency. Therefore, these data are helpful in investigating the linkages between public spending, public governance, and economic growth in the provinces. In this study, public spending is decomposed into public investment, current expenditure, and expenditures on education, health, and science. We use the Vietnam Provincial Competitiveness Index (PCI) as a proxy for economic governance, and PCI sub-indices, including informal charges, transparency, and policy bias, are used to estimate the linkages among public spending, private investment, and growth, on the assumption that public governance improves the growth effect of the interaction term between public spending and private investment. Regarding the conditional relationship between economic governance and public services, Malesky et al. (2007) confirmed that infrastructure development and human capital accumulation are significantly linked to PCI scores. This means that provinces with a higher PCI score (i.e., better governance) can create higher living standards at the same development level.

Most cross-country studies on public spending, public governance, and economic growth in the developing world contribute to the literature on how public governance conditions the efficiency of the interaction term between public spending and private investment in these countries. We add to the literature by providing evidence from a subnational study of public spending, public governance, and economic growth in Vietnamese provinces. This study makes two contributions to the literature. First, by enriching the understanding of the combined factors of local public spending, public governance, and growth performance, it could help Vietnam's local governments provide better public welfare. Second, the study implies that local public governance plays a critical role in improving the growth effect of the interaction between public spending and private investment across Vietnamese provinces. Local governments cannot wait for greater economic-social development before they improve the quality of institutions and governance. Instead, reduced informal charges, unbiased public policy, and enhanced transparency are imperative for private sector development and local economic growth. To optimize fiscal policy and decentralization, the central government needs to recognize that economic governance has an impact on local government efficiency.

The rest of this paper is organized as follows. Section 2 reflects briefly on the prevailing context of Vietnam in line with the topic addressed. Section 3 presents some of the previous studies on the relationship among public spending, governance, private investment, and economic growth. Key aspects of the empirical models, together with data sources, are outlined in Section 4. The empirical

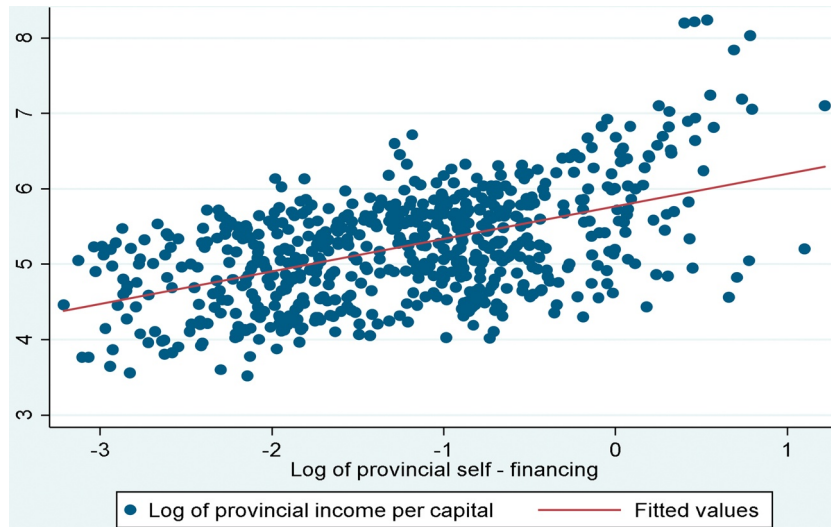


Fig. 1. The relationship between provincial self-financing and provincial income per capita over the 2006-2015 period.
 Source: Data from the Statistical Yearbook (Provincial Bureau of Statistics), and authors' calculation.

results of the model are discussed in Section 5. Some conclusions following the analysis are presented in Section 6.

2. Overview of Public Governance, Public Spending, Private Investment, and Growth at the Provincial Level in Vietnam

Since the social and economic reforms initiated in the 1980s (called *doi moi*, or renovation), Vietnam's economy has been in the process of transitioning from a command to a market-oriented economy, fully integrated into the global economy. Thanks to sustained high economic growth rates during the renovation, Vietnam has advanced from being a desperately poor country to a vibrant emerging market in the 2000s. One important pillar of the reform is decentralization, in which larger social-economic responsibilities are devolved to local governments, with the implication of greater competition among provinces. As a result, the capacity of local governments for self-financing has been significantly improved through increases in decentralized revenue to fund local expenditures. As illustrated in Fig. 1, which displays the relationship between provincial self-financing and provincial income per capita growth over the period 2006–2015, this self-financing is calculated as the ratio of decentralized revenue to total local expenditures.

In addition to growth in income per capita, local public spending on infrastructure, education, health care, and business services has made a critical contribution to local economic growth. Without these spending items, such rapid income growth among provinces would have been impossible. Remarkably, public spending policy has changed since the 2008 financial crisis. For example, the



Fig. 2. Provincial fiscal transfers and public investment size over the period 2006-2015.
 Source: Data from the Statistical Yearbook (Provincial Bureau of Statistics), and authors' calculation.

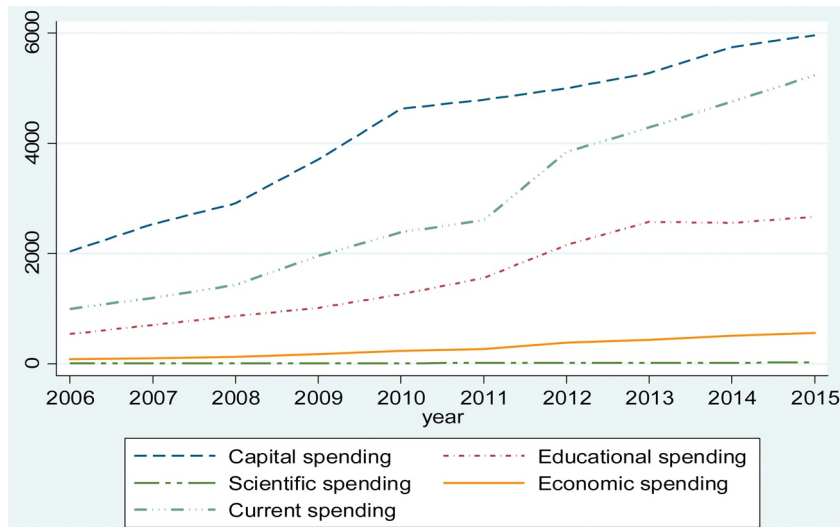


Fig. 3. Mean value of provincial public expenditures per capita (VND Million) in the period 2006-2015.

Source: Data from the Statistical Yearbook (Provincial Bureau of Statistics), and authors' calculation.

central government promotes fiscal decentralization and increases autonomy, local responsibility, and transparency by local governments. As a result, the size of local public investment and fiscal transfers is lower (see Fig. 2). Moreover, the structure of local public spending has been adjusted to meet the United Nations millennium development goals. The structure of public spending prioritizes human resources, education, health, science, and technology.

Fig. 3 shows the trends in disaggregated public expenditures at the local level. The relative contribution of government spending needs to be assessed for two reasons. First, the link between public spending and economic growth is still uncertain. Nguyen and Anwar (2011) showed that local spending has a negative effect on provincial economic growth, while Nguyen-Van et al. (2018) did not find any evidence to support the significant effects of public spending on economic growth in the provinces of Vietnam. As suggested by Bayraktar and Moreno-Dodson (2015), the growth effects of public spending and its components are available only in countries or regions with growth dynamics, macroeconomic stability, and an aggressive private sector. Second, few studies have paid attention to the components of public spending in terms of economic growth in Vietnam. Public sector reforms can require local governments to do more with less. Therefore, information on the relative contributions of various types of public spending is helpful to enable the allocation of public resources to be better targeted in line with the development goals.

Under decentralization, a PCI was constructed to measure and rank the capacity of economic governance in association with local private sector development. The PCI was first introduced in 2005 to rank forty-seven provinces. Since 2006, sixty-three provinces in Vietnam have been included in the ranking list. Fig. 4 depicts the shifts in overall PCI scores and provincial income per capita growth

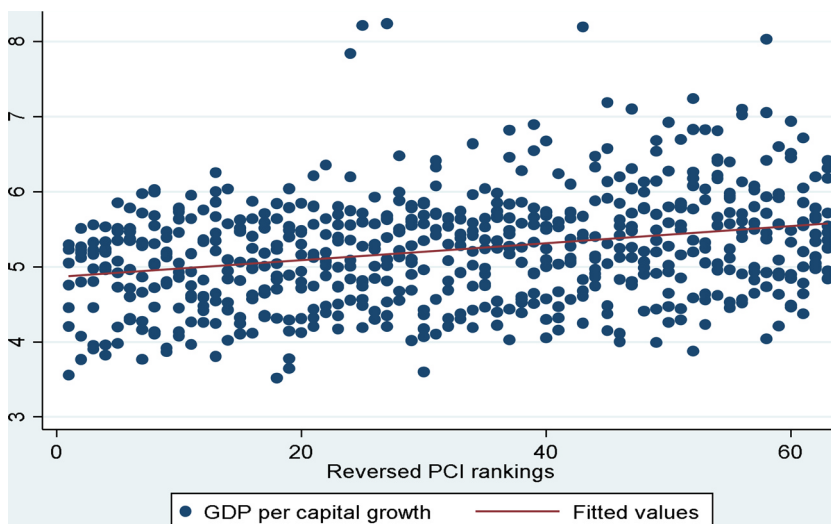


Fig. 4. The relationship between reversed PCI rankings and provincial income per capita over the period 2006-201.

Source: Data from the Statistical Yearbook (Provincial Bureau of Statistics) and VCCI, and authors' calculation.

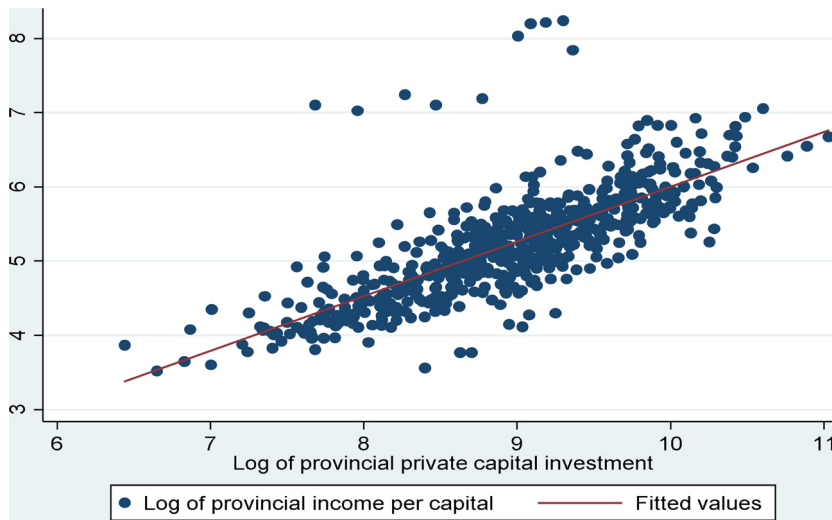


Fig. 5. The relationship between provincial income per capita and private capital investment over the period 2006–2015.
Source: Data from the Statistical Yearbook (Provincial Bureau of Statistics), and authors' calculation.

over the period 2006–2015.

Since the period of economic reform, in addition to the public sector, the private sector has been recognized as a formal sector, becoming a focus of development via several critical measures implemented by Vietnam's government. A business environment and a legal framework for private enterprise have gradually come into force through the promulgation of business laws and regulations. Additionally, during the 1990s public investment in infrastructure grew rapidly to support the phenomenal growth of the private sector.

Fig. 5 shows the growth trends in domestic private capital investment per capita and income per capita at the provincial level. Some arguments show that infrastructure constraints in Vietnam have yet to be addressed. Infrastructure bottlenecks are considered the biggest hindrance to the business environment in Vietnam (Thanh and Dapice, 2009). Therefore, it is unclear whether public investment crowds in/out private investment among Vietnam's provinces.

3. Literature review

The connections among governance, public spending, private investment, and economic growth have been investigated in several ways, which are relevant to the model developed in the following sections.

First, there is the question of the linkages between public spending and economic growth. According to endogenous growth theory, the existence of public goods and services generates positive spillover effects on production or utility (Barro, 1990; Cazzavillan, 1996). The complementarity between public expenditure and private consumption increases economies of scale, which are sufficiently strong to produce increasing returns in the utility function of representative agents, thus generating economic conditions that increase growth in the gross domestic product (GDP) (Cazzavillan, 1996). However, it is argued that the direct linkage between public spending and outcomes can be uncompromising < pls clarify > because of the potential crowding-out effect of public provision on private sector provision. The line of reasoning is that the efficiency of public spending depends on the quality of budgetary institutions. Instead, governments need to substitute public spending for private investment to improve the marginal effect on outcomes (Curristine et al., 2007; Rajkumar and Swaroop, 2008). Many empirical studies, using a variety of datasets and econometric methods, have failed to reach unambiguous conclusions: some reported positive linkages, while others found negative relationships (Carboni and Medda, 2011; Carrère and de Melo, 2012; Daniel and Roşoiu, 2015; Dissou et al., 2016; Kiran, 2014; Romero-Avila and Strauch, 2008; Samudram et al., 2009). Some investigations have detected nonlinear or threshold effects of public expenditure on economic growth, suggesting that public spending might promote economic growth performance up to a particular level but that further increases in public spending above an “optimal” threshold might be detrimental to economic growth (Afonso and Furceri, 2010; Cavicchioli and Pistoiesi, 2016; Chen and Lee, 2005).

The second set of questions relates to how the quality of public governance affects the relationship between public spending and growth. According to institutional theory, institutions affect market behaviors and resource allocation because of asymmetric information problems, transaction costs, and risks (Acemoglu and Robinson, 2010). The quality of institutions contributes significantly to connections between production inputs and economic development (Jones and Romer, 2010) and public spending and outcomes (Curristine et al., 2007; Rajkumar and Swaroop, 2008). Using a variety of indicators to capture different aspects of public governance, recent studies have confirmed significant connections between governance and growth (Gerry et al., 2010; Markus and Mendelski, 2015; Rajkumar and Swaroop, 2008; Wilson, 2016). For example, Rajkumar and Swaroop (2008) suggested that differences in the efficacy of public spending in improving human development outcomes can be explained by the quality of governance, as measured

by the level of corruption and the quality of bureaucracy. Darden (2002) indicated that corruption is related to graft and informal mechanisms of control by state authorities. As a result, corruption can increase transaction costs and resource misallocation. Whitsel (2011) showed that informal charges and corruption lead to increases in the cost of education and thus reduce human capital accumulation in Tajikistan. Therefore, combating corruption conditions the efficiency of public spending and economic growth (d'Agostino et al., 2016). Importantly, the quality of governance shapes the linkage between corruption and growth (Dzhumashev, 2014). In an analysis of interactions among corruption, taxation, and economic growth, Attila (2008) reached conclusions that were consistent with the existence of a nonlinear relationship between public expenditures and economic growth, thus indicating that, although under certain conditions corruption appeared to affect growth rates positively, its negative impact on fiscal revenue offset any possible positive effect of corruption on growth. Cooray (2009)'s extensive empirical study on seventy-one countries found that both the size and quality of government activity are important for economic growth, concluding that to achieve improved growth performance, investment in the capacity of enhanced governance should be a priority.

The third set of questions that is relevant to how public governance affects the linkage between public spending and private sector investment is related to the fact that whether public investment complements or crowds out private investment has different effects on economic performance (Erden and Holcombe, 2005). It has generally been concluded in empirical studies that real GDP and growth in income per capita are positively related to the growth rate of private sector investment (Mallick, 2013; Morrissey and Udomkerdmongkol, 2012; Oshikoya, 1994; Sineviciene and Railiene, 2015). Since the emergence of endogenous growth theory < when? >, the fundamental influence of government policy on the long-run growth performance of countries has been widely recognized in the economic growth literature (Tanzi and Zee, 1997). This is particularly the case with respect to government activities related to infrastructure development, human capital accumulation, regulation of international trade and investment, maintenance of institutional quality, development of the financial market, and the conduct of fiscal policy, which are critical for economic activities and private sector development. However, the relationship between public spending and private investment remains somewhat controversial, with some continuing debate as to whether public spending and private investment are complementary or competing. Public investment that increases the marginal productivity of private capital is complementary to or crowds in private investment (Morozumi and Veiga, 2016). However, public investment reduces the return on private capital and then crowds out private investment when they compete for the same resources (Abel, 2017).

Importantly, weak institutions are associated with biased public policy through limited access to financial services, thus leading to reducing the positive effects of public investment and enlarging the crowding out of private sector investment. More government effectiveness helps build trust with transparency and unbiased public policies and provides rules and stability that are necessary for firms to plan investment and expand their businesses in the long run, creating productive interaction between government, public agents, and firms (Kousky et al., 2006). More generally, many studies have agreed that transparency conditions government effectiveness and positive effects of public spending on economic performance (López and Fontaine, 2019; Montes et al., 2019; Morozumi and Veiga, 2016; Phuc Canh, 2018 < see query in ref list >). Moreover, transparency and accountability in government policy making are important in achieving and maintaining growth in long-run private sector investment (Braga Tadeu and Moreira Silva, 2013). Weak governance, however, leads to increased risk related to private sector investment decisions. In this regard, Wang et al. (2019) indicate that less risk could attract more private sector investment. Therefore, high public governance levels in terms of less corruption, government effectiveness, regulatory quality, and rule of law reduce the negative effect of risk on private investment and thus increase the crowding-in effect of public investment on private investment.

This review of the literature leads to the conclusion that good governance conditions the crowding-in effect of public spending on private investment and thus the positive effect of public spending on outcomes. Good governance requires governments to enhance the level of transparency and accountability, which is necessary to control corruption and reduce informal charges. Moreover, unbiased public policy is also a necessity for good governance, which is more effective for improving the growth effect of the interaction between public spending and private investment. In the same vein, regarding the context of a country or subcountry, Corrado and Rossetti (2018) found that corruption in local public spending is associated with local public administration. Interestingly, enhancing social capital and transparency is critical for reducing corruption and crime rates across regions in Italy. Zakharov (2019) found that, under the effect of local government corruption, both public sector investment and private investment decline and thus induce lower growth in Russian regions. Gao (2019) indicated that easy access to land and productive public investment drive private sector investment in improving economic growth in Chinese cities. Li and Zhang (2018) emphasized that transparency by local authorities with fewer factional ties reduce tax burdens per capita and provide better public goods/services for residents in Chinese provinces.

To date, very few empirical studies have been conducted at the provincial level in Vietnam on the effects of public spending and public governance on economic growth (for a study on aggregate data, see, e.g., see query in ref list > Nguyen Canh, 2018). Nguyen et al. (2019) found that provincial governance and public administration measured by transparency, accountability, and delivery of public services significantly affect economic growth and poverty reduction in Vietnamese provinces. Before this study, using the PCI as a proxy for local economic governance, Su and Bui (2017) showed that aspects of local governance, such as the legal environment, transparency, and access to information, improve the effectiveness of public spending on private investment across Vietnamese provinces. Tran et al. (2009) indicated that improvement in providing market information, land access, and labor training has a positive influence on firm performance, while defectiveness in the judiciary system and administrative services were detrimental to performance. However, these studies overlook regional patterns of public spending and local governance, but Nguyen-Van et al. (2018) showed that local public spending does not significantly affect economic growth, and human capital has a significant effect on productivity growth in Vietnamese provinces. Moreover, compared to the results of these studies, the significance of economic governance in improving outcomes remains somewhat ambiguous in Vietnam. McCulloch et al. (2013) found few significant relations

between most aspects of PCI and private investment. Therefore, given the conflicting and incomplete findings reported in earlier studies, the linkage between economic governance and private sector investment need to be further investigated in a systematic way, particularly how the latter interacts with government expenditure to promote economic growth. This study looks at the relationships between the performance of private sector firms and aspects of local public governance, such as informal charges, transparency, and biased policy.

4. Research model, data sources, and estimation techniques

4.1. Research model

The empirical model developed in this section has as its starting point the production function of the province:

$$Y_{it} = A_{it} K_{it}^{\alpha} GINV_t^{\beta} \quad (1)$$

where $i = 1, 2, 3, 4 \dots N$ (province); $t = 1, 2, 3, 4, \dots, T$ (time); Y is provincial output per capita; K is provincial private capital stock per capita; $GINV$ is central government investment per capita, which reflects the role of the central government in provincial economic growth; and A measures TFP. We assume that $\alpha + \beta \neq 1$, that is, returns are not constant to the scale production function.

From the endogeneity theory perspective, A_{it} is decided by factors in the structure of growth model, which is related to policy levers controlled by local authorities. In this study, the factors included that can affect long-run growth at the provincial level are self-financing, public spending, governance, and the central fiscal transfers to provinces. First, self-financing refers to the level of freedom by local governments in allocating fiscal revenues, which is calculated by the ratio of total decentralized revenue to total provincial government expenditure. As in [Song et al. \(2018\)](#), this indicator is used as a proxy for fiscal decentralization. If fiscal decentralization is high, the local authorities' social and economic development policies are indicated by self-financing efforts to fund local public expenditures. Second, the allocation of local public expenditures is indicated by improving public services/goods provision and the quality of local governance. Public spending on basic public services (e.g., infrastructure, education, health) is assumed to affect productivity growth and economic growth. Conversely, inadequate infrastructure and inefficient public spending can be detrimental to productivity growth, thus leading to less economic growth. Second, public governance determines the efficiency of public resource allocation and productivity growth. More specifically, good governance can: (1) drive productive intensive activities by the private sector, (2) allocate resources among firms and economic sectors efficiently, and (3) create favorable opportunities for the use of advantageous technologies. Therefore, good governance, characterized by lower corruption, transparency, and greater accountability, is assumed to improve the efficiency of public spending ([López and Fontaine, 2019](#); [Montes et al., 2019](#); [Morozumi and Veiga, 2016](#)) and thus private investment development ([Braga Tadeu and Moreira Silva, 2013](#); [Wang et al., 2019](#)). Third, fiscal transfers means the central fiscal transfers to local governments to address the fiscal gap across provinces. The literature on fiscal decentralization emphasizes that these fiscal transfers lead to a common pool problem and drive local authorities to disregard budget constraints and fiscal competition, thus inducing poor provision of public goods ([Tiebout, 1956](#)). This means that the effect of fiscal transfers on productivity growth is the opposite of that of self-financing. This needs be tested to clarify whether a tradeoff exists between self-financing and fiscal transfers in affecting provincial productivity growth.

A_{it} is now specified by:

$$A_{it} = A_{i0} SF_{it}^{\sigma} LGS_{it}^{\omega} GOV_{it}^{\varphi} FT_{it}^{\gamma} \quad (2)$$

where SF is the provincial self-financing level, as a proxy for fiscal decentralization; LGS is provincial public spending, GOV is provincial economic governance, as a proxy for local institutions, and FT is fiscal transfers; σ , ω , φ , and γ measure the efficiency of self-financing, public spending, economic governance, and fiscal transfers with respect to productivity growth.

Inserting Equation (2) into Equation (1), we obtain:

$$Y_{it} = A_{i0} K_{it}^{\alpha} GINV_t^{\beta} SF_{it}^{\sigma} FT_{it}^{\gamma} LGS_{it}^{\omega} GOV_{it}^{\varphi} \quad (3)$$

Taking the natural logarithm of Equation (3), we obtain the following equation for output growth in the province:

$$\ln Y_{it} = \ln A_{i0} + \alpha \ln K_{it} + \beta \ln GINV_{it} + \sigma \ln SF_{it} + \gamma \ln FT_{it} + \omega \ln LGS_{it} + \varphi \ln GOV_{it} + \xi_{it} \quad (4)$$

Built on the relationships derived in Equation (4), the following dynamic regression equations are used to analyze the relationship among public spending, economic governance, and economic growth in Vietnamese provinces:

$$\Delta \ln Y_{it} = \beta_0 + \beta_1 \ln Y_{it-1} + \beta_2 \ln K_{it} + \beta_3 \ln GINV_{it} + \beta_4 \ln SF_{it} + \beta_5 \ln FT_{it} + \beta_6 \ln LGS_{it} + \beta_7 \ln GOV_{it} + \beta_8 \text{CONTROL}_{it} + (\gamma_i + \eta_t + \varepsilon_{it}) \quad (5)$$

where, $i = 1, 2, 3, \dots, N$; N denotes the province; $t = 1, 2, 3, \dots, T$; T is for the time; γ_i is a vector of province-specific fixed effects, η_t is a vector of time effects, ε_{it} is the error term, $\varepsilon_{it} \approx i. i. d(0, \sigma)$; $\Delta \ln Y_{it}$ is a proxy for economic growth; CONTROL is a control variable.

4.2. Measures of variables and method

4.2.1. Dependent variable

- Y is the gross provincial product (GPP) per capita adjusted by the consumer price index (CPI). $\Delta \ln Y_{it}$ is a proxy for provincial real economic growth.

4.2.2. Independent variables

- K is the provincial private capital stock per capita adjusted by the CPI. $K_{it} = K_{it}^{DPRI} + K_{it}^{FDI}$, in which $K_{it}^{DPRI} = I_{it}^{DPRI} + (1 - \delta)K_{it-1}^{DPRI}$ and $K_{it}^{FDI} = I_{it}^{FDI} + (1 - \delta)K_{it-1}^{FDI}$; I_{it}^{DPRI} and I_{it}^{FDI} are the inflows of annual provincial private capital investment, including provincial domestic private investment (DPRI) and foreign direct investment (FDI); δ is the depreciation rate, which is assumed to be a constant 0.05, as in [Mankiw et al. \(1992\)](#).
- $GINV$ is the central government investment spending per capita on roads, highways, and airports, which is used to estimate the external effect of central government investment on provincial economic growth.
- SF is the provincial self-financing level, as a proxy for fiscal decentralization, which is calculated by:

$$FT = \frac{\text{total provincial decentralized revenue}}{\text{Total provincial government expenditure}} \times 100$$

where total provincial decentralized revenue includes revenue that is 100% retained and revenue that is shared between the provincial government and the central government.

- FT is central fiscal transfers to local governments, which are calculated as

$$FT = \frac{\text{Provincial fiscal transfers}}{\text{Total provincial government expenditure} + \text{fiscal transfers}} \times 100$$

- LGS is provincial government expenditures per capita adjusted by provincial inflation, which are disaggregated into the following expenditure items:
 - o Provincial public investment (PUBINV), which corresponds to investment in assets and infrastructure.
 - o Provincial current spending (CUR), which refers to expenditures on salary, goods, and public services.
 - o Provincial education spending (EDU), which denotes expenditures on salaries, school equipment, assets, goods, and services for the local education system.
 - o Provincial economic spending (ECO), which refers to expenditures on irrigation and in agriculture, forestry, and fishing.
 - o Provincial government spending on science and technology (SCIEN).
- $CONTROL$ is a dummy variable for the 2008 financial crisis, which takes a value 0 for before the 2008 financial crisis and 1 after the 2008 financial crisis.

4.2.3. Measurement of public governance

In Equation (5), GOV is economic governance proxied by the provincial competitiveness index (PCI). According to [Malesky et al. \(2018\)](#), the PCI is recalibrated every four years to ensure that it keeps up with economic reforms and the policy objectives of the central and local governments. Key methodological changes were made in 2006, 2009, 2013, and 2017, with the PCI's sub-indices dropped, changed, or merged in the weighting scheme. These changes in the overall score make it difficult to know whether changes in scores reflect improvement in governance. In this study, first, an alternative approach is to use the PCI ranking, which is not subject to the same change in measurement as the general PCI scores. PCI rankings demonstrate provincial economic performance, from first to sixty-third place, in which the province that is ranked first (last) has the strongest (poorest) economic performance. However, a better estimation procedure is to reverse the order of the PCI rankings to estimate the ascending effect of provincial economic performance on growth. This implies that provinces with a higher reverse PCI ranking have better performance and thus are related to a higher growth effect from economic performance.

Second, to understand how public governance conditions the growth effects of public spending and private investment, we use three PCI sub-indices, including informal charges, transparency, and policy bias as a proxy for good public governance (for more detail, see Appendix A.2). In this respect, [Malesky et al. \(2015\)](#) found that overall governance does not predict investment, whereas the sub-index of transparency has a robust and significant impact on both foreign and domestic investment in Vietnamese provinces. Regarding good governance, [Nguyen et al. \(2015\)](#) showed that transparency and greater accountability are crucial for combating corruption and efficiency in government expenditures on education and health care in Vietnamese provinces. In addition, good governance builds trust in unbiased public policies and provide stable rules that are necessary for firms to plan investment and expand their business ([Kousky et al., 2006](#)).

Equation (5) can be estimated by pooled ordinary least squares and fixed effects (FE) with robust standard errors. However, the traditional FE estimator is inconsistent because the sample mean of the lagged dependent variable is correlated with that of the idiosyncratic error term, an issue that becomes particularly significant in dynamic panel-data models that have relatively few time periods and $<$ meaning a large sample size? $>$ many individual units ([Nickell 1981](#)). The inclusion of the lagged dependent variable

in Equation (5) introduces endogeneity directly into the model under consideration. In the context of a dynamic panel model such as Equation (5), generalized method of moments (GMM) estimators are often implemented. Moreover, Bond et al. (2001) indicated that empirical economic growth models are preferable to the system-GMM estimator developed by Arellano and Bover (1995) and Blundell and Bond (1998) because the additional instruments can be valid under restraints on the initial conditions, which may be consistent with economic growth models.

It is notable that GMM estimators can eliminate all time-invariant variables because of a first-difference transformation. Following Kripfganz and Schwarz (2015), we apply a sequential (two-stage) estimation. In the first stage, we regress Equation (5) with the two-step system-GMM estimator with corrected robust standard errors, as in Windmeijer (2005).

$$Y_{it} = \beta X'_{it} + \bar{u}_i + \varepsilon_{it} \text{ with } \bar{u}_i = f'_i + u_i \quad (6a)$$

where X'_{it} is a vector of explanatory variables as in Equation (5), and f'_i corresponds to time-invariant variables reflecting local fixed characteristics. We know that economic growth depends on many factors other than fiscal decentralization and public governance. To control for these influences, we take into account the local social and economic geographic factors. In the second stage, the coefficients of time-invariant variables are estimated by using a two-step IV estimation for the residuals from the system-GMM estimator.

$$Y_{it} - \hat{\beta} X'_{it} = f'_i + u_i + \varepsilon_{it} \text{ with } \hat{\varepsilon}_{it} = \varepsilon_{it} - X'_{it}(\beta - \hat{\beta}) \quad (6b)$$

To define time-invariant variables, we select the Southeast and Mekong River Delta region with high PCI scores as the basic group. Three regional dummies are used to represent the Northern Mountain region, the Red River Delta region, and the Central and Highlands region. In Equations (6a) and (6b), the reliability of instruments in the system-GMM is tested through the application of the Hansen-J statistic, with the p -value of the Hansen statistic, and for the instruments to be tested. This means that the Hansen-J test is implemented for Equations (6a) and (6b), respectively. It is also necessary to conduct the Arellano-Bond test of autocorrelation with the test for AR(2), which is of particular importance as it detects autocorrelation in levels.

4.3. Description of data

The source of data in this study is the *Statistical Yearbook* (Provincial Bureau of Statistics).¹ The PCI and data on central government investment are from the Vietnam Chamber of Commerce and Industry and < is this a publication? > Key Indicators for Asia and the Pacific 2017 (ADB), respectively. The panel dataset covers sixty-two of the sixty-three Vietnamese provinces over the period 2006–2015. DakNong Province is omitted from the sample because of missing data. The definition, calculation, descriptive statistics, and sources of the variables are listed in Table 1. To reduce heteroskedasticity, we transform the variables in Table 1 into their natural logarithm.

The correlation coefficients between variables are reported in Table 2, which shows that the correlation coefficients of *EDU*, *ECO*, and *CURR* are very high (over 0.800). To avoid collinearity, these variables are separated when their effects on economic growth are estimated. Other correlation coefficients are acceptable for incorporation into the empirical models.

5. Estimation results

5.1. Public spending and economic growth

First, the question of how public spending components have significant impacts on economic growth in a province is answered in Table 3. The results are estimated with a sequential (two-stage) estimation. The test for AR(2) and Hansen-J tests (1) for the first-stage estimation and Hansen-J tests (2) for the second-stage estimation show that the system-GMM estimators do not have significant issues related to autocorrelation and that the instruments are valid.

As shown in Table 3, Model (1) shows only private investment (see Column 1). Model (2) is estimated with central public investment (*GINV*), self-financing (*SF*), fiscal transfers (*FT*), provincial public spending (*PUBINV*), science expenditure (*SCIEN*), and economic expenditure (*ECO*) (see Column 2). Model (3) is adjusted by adding education expenditure (*EDU*) (see Column 3), and Model (4) is modified by incorporating current spending (*CUR*) (see Column 4).

The estimated results show that the coefficient of the self-financing level is positive, whereas that of fiscal transfers is negative in relation to provincial economic growth, suggesting that a significant tradeoff exists between the self-financing level (or fiscal decentralization) and fiscal transfers in promoting provincial economic growth in Vietnam. This result implies that the central government should increase the degree of local governments' fiscal freedom, which enables the governments collecting local revenues to meet the needs for public goods provision. At the same time, the central government should limit fiscal transfers to shift the prevailing mode to one of hard budget constraints, thus boosting local economic performance. Hence, discretion by the central government through fiscal transfers should be reduced to foster fiscal discipline across provinces. The coefficients of public investment and science expenditure are positive, suggesting that provincial public investment and science expenditure make a positive contribution to provincial economic growth. This finding varies from that of Nguyen-Van et al. (2018), which indicated that the effect of local public spending on the economic growth of provinces is insignificant.

¹ The General Statistics Office of Vietnam (GSO) helped us to obtain these data.

Table 1
Definitions and descriptive statistics of the variables.

Variables	Definition, description, and source	Obs.	Mean	Std.	Min.	Max.
Provincial output per capita (<i>Y</i>)	Gross provincial nominal product (GPP) per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	25828.56	33993.02	3379.26	378734.2
Provincial private investment per capita (<i>PRINV</i>)	Provincial domestic private investment per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	5066.753	3903.092	259.373	34245.48
Provincial foreign direct investment (<i>FDI</i>)	Provincial foreign direct investment per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	19.020	52.461	0	579.943
Provincial capital stock (<i>K</i>)	$K_{it} = K_{it}^{DPR} + K_{it}^{FDI}$ in which $K_{it}^{DPR} = I_{it}^{DPR} + (1 - \delta)K_{it-1}^{DPR}$ and $K_{it}^{FDI} = I_{it}^{FDI} + (1 - \delta)K_{it-1}^{FDI}$; I is the inflows of domestic private capital investment (DPR) and foreign direct investment (FDI), respectively; δ is the depreciation rate, which is assumed to be constant, 0.05	620	10027.39	7498.835	62.707	61801.63
Central government investment (<i>GINV</i>)	Central government investment per capita (VND Million), from Key Indicators for Asia and the Pacific 2017, ADB	620	2167.217	685.269	1066.276	3028.425
Provincial Self – Financing (<i>SF</i>)	Measured by total provincial decentralized revenue to total provincial government expenditure, from the annual provincial Statistical Yearbook	620	0.423	0.415	0.040	3.383
Provincial Fiscal Transfers (<i>FT</i>)	Measured by total provincial fiscal transfers to total provincial government expenditure plus fiscal transfers, from the annual provincial Statistical Yearbook	620	24.985	11.723	0.857	48.547
Provincial public investment per capita (<i>PUBINV</i>)	Provincial public investment per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	3922.09	3140.76	480.137	25419.4
Provincial education spending per capita (<i>EDU</i>)	Provincial education spending per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	1154.608	939.216	167.528	12802.05
Provincial science and research spending per capita (<i>SCIENCE</i>)	Provincial science and research spending per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	14.076	9.051	1.952	93.870
Provincial economic services spending per capita (<i>ECO</i>)	Provincial economic services spending per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	620	287.262	235.093	15.611	1599.397
Provincial current spending per capita (<i>CUR</i>)	Provincial current spending per capita adjusted by provincial inflation (VND Million), from the annual provincial Statistical Yearbook	615	2880.265	1921.844	541.957	11232.69
Provincial public governance (<i>GOV</i>)	Provincial Competitiveness Index (PCI) ranking as a proxy of provincial public governance from Vietnam's VCCI	620	32.398	38.013	1.000	63.00
Informal charges (<i>CHARGE</i>)	Informal charges scores, Provincial Competitiveness Index (PCI) from Vietnam's VCCI	620	6.210	1.003	2.810	8.940
Transparency (<i>TRANS</i>)	Transparency scores, Provincial Competitiveness Index (PCI) from Vietnam's VCCI	620	5.864	0.830	2.460	8.850
Policy bias (<i>BIAS</i>)	Policy bias scores, Provincial Competitiveness Index (PCI) from Vietnam's VCCI	620	5.445	1.419	1.750	8.770

Table 2

Correlation matrix of variables.

Sources: Statistics are over the period of 2006–2015. Ln (Y): log of gross provincial nominal product per capita adjusted by inflation. Ln (K): log of provincial capital stock per capita adjusted by inflation. Ln (GINV): log of central government investment per capita. Ln (SF): log of provincial government decentralized revenue to total provincial government expenditure. Ln (FT): log of provincial fiscal transfers to total provincial government expenditure and fiscal transfers. Ln (PUBINV): log of provincial public investment per capita adjusted by inflation. Ln (EDU): log of provincial education spending per capita adjusted by inflation. Ln (SCIEN): log of provincial scientific research spending per capita adjusted by inflation. Ln (ECO): log of provincial economic services spending per capita adjusted by inflation. Ln (CUR): log of provincial current spending per capita adjusted by inflation; Ln (GOV): log of Provincial Competitiveness Index (PCI) ranking as a proxy of public governance.

	$\Delta \ln(Y)$	$\ln(K)$	$\ln(GINV)$	$\ln(SF)$	$\ln(FT)$	$\ln(PUBINV)$	$\ln(EDU)$	$\ln(SCIEN)$	$\ln(ECO)$	$\ln(CUR)$	$\ln(GOV)$
$\Delta \ln(Y)$	1.000										
$\ln(K)$	−0.111***	1.000									
	0.006										
$\ln(GINV)$	−0.079**	0.693***	1.000								
	0.050	0.000									
$\ln(SF)$	−0.040	0.336***	−0.002	1.000							
	0.324	0.000	0.953								
$\ln(FT)$	0.057	−0.402***	−0.077*	−0.681***	1.000						
	0.159	0.000	0.055	0.000							
$\ln(PUBINV)$	−0.024	0.484***	0.504***	0.001	−0.218***	1.000					
	0.558	0.000	0.000	0.974	0.000						
$\ln(EDU)$	−0.146***	0.444***	0.778***	−0.341***	0.146***	0.554***	1.000				
	0.000	0.000	0.000	0.000	0.000	0.000					
$\ln(SCIEN)$	−0.111***	0.561***	0.579***	0.082**	−0.294***	0.593***	0.624***	1.000			
	0.006	0.000	0.000	0.043	0.000	0.000	0.000				
$\ln(ECO)$	−0.136***	0.548***	0.767***	−0.151***	−0.059	0.627***	0.867***	0.677***	1.000		
	0.001	0.000	0.000	0.000	0.141	0.000	0.000	0.000			
$\ln(CUR)$	−0.123***	0.554***	0.806***	−0.236***	0.017	0.657***	0.949***	0.696***	0.909***	1.000	
	0.002	0.000	0.000	0.672	0.000	0.000	0.000	0.000			
$\ln(GOV)$	0.008	0.171***	−0.011	0.472***	−0.356***	−0.056	−0.261***	−0.098**	−0.167***	−0.199***	1.000
	0.841	0.000	0.780	0.000	0.000	0.165	0.000	0.015	0.000	0.000	

(***), (**), (*) denote significance at a level of 1%, 5%, and 10%, respectively.

The coefficients of other types of expenditure are not as expected. The coefficients of *CUR*, *EDU*, and *ECO* are insignificant, suggesting that expenditures on education, business services, and public administration are inefficient, thereby limiting economic growth in provinces. It is not surprising that private capital stock has a significantly negative effect on economic growth because Vietnam's physical capital accumulation may have begun to have diminishing returns, which is consistent with Jin and Zou (2005), who examined China. Nguyen and van Dijk (2012) showed that in Vietnam the business environment and institutions are detrimental to private sector investment. This suggests that policy makers should develop strategies to create a favorable business environment and then raise productivity growth in private capital investment. This requires better governance to diminish transaction costs and risks.

5.2. Public spending, private investment, and economic growth

By matching public spending components with private investment, we create interaction variables such as $\ln PUBINV * \ln K$, $\ln CUR * \ln K$, $\ln ECO * \ln K$, $\ln SCIEN * \ln K$, and $\ln EDU * \ln K$. Private investment is related to expenditures on public investment, economic services, science and technology, and public administration. The purpose of including the interaction terms is to examine whether increased public spending leads to improvement in the effect of private investment on economic growth. The estimated results are presented in Table 4 with Models (1)–(5).

The table shows that adding the interaction terms change the sign and value of $\ln PUBINV$, $\ln CUR$, $\ln ECO$, $\ln SCIEN$, $\ln EDU$, and $\ln K$. Specifically, coefficients on $\ln PUBINV * \ln K$, $\ln CUR * \ln K$, $\ln ECO * \ln K$, $\ln SCIEN * \ln K$, and $\ln EDU * \ln K$ are all negative in Models (1)–(5), suggesting that an increase in public spending reduces the effect of private investment on economic growth in the provinces. This finding implies that public spending and private investment may still be substitutes. Public spending is not expected to raise the effectiveness of private investment at the provincial level, so it is necessary to further investigate the role of public government in modifying the growth effect of interaction between public expenditure and private investment.

5.3. Economic governance and economic growth

Malesky et al. (2017) emphasized that strong evidence indicates that improvements in economic governance are associated with private sector development at the provincial level. In this section, the focus is on testing the effect on economic growth of economic governance, as reflected in the PCI. This analysis uses the mean of the reverse PCI rankings of the full sample as a fundamental criterion to determine high and low levels of economic governance. As shown in Table 1, based on the mean PCI ranking, we define ranking thirty-second to sixty-third in the reverse PCI rankings as a high PCI ranking for provinces and ranking first to thirty-first as a

Table 3Impacts of public spending on provincial economic growth: A disaggregate analysis. Dependent variable: Provincial economic growth, $\Delta \ln(Y)$.

VARIABLES	(1)	(2)	(3)	(4)
The first stage estimation				
Lag of $\ln(Y)$	−0.041*** [0.010]	−0.171*** [0.019]	−0.151*** [0.020]	−0.169*** [0.026]
$\ln(K)$	−0.047** [0.021]	−0.074** [0.032]	−0.067** [0.027]	−0.082*** [0.029]
$\ln(GINV)$		0.174*** [0.036]	0.171*** [0.039]	0.124*** [0.035]
$\ln(SF)$		0.056*** [0.014]	0.049*** [0.014]	0.067*** [0.017]
$\ln(FT)$		−0.057*** [0.017]	−0.040** [0.016]	−0.057*** [0.019]
$\ln(PUBINV)$		0.033** [0.013]	0.036*** [0.012]	0.030* [0.016]
$\ln(SCIEN)$		0.024* [0.014]	0.032*** [0.012]	
$\ln(ECO)$		0.010 [0.014]		
$\ln(EDU)$			−0.019 [0.023]	
$\ln(CUR)$				0.052 [0.036]
Crisis 2008	0.081*** [0.015]	0.054*** [0.016]	0.055*** [0.017]	0.070*** [0.017]
The second stage estimation				
Northern Mountain Region	−0.007 [0.015]	0.004 [0.011]	0.013 [0.010]	−0.002 [0.011]
Red River Delta Region	0.053*** [0.021]	0.031 [0.020]	0.034* [0.019]	0.034* [0.020]
Highlands and Central Region	0.004 [0.013]	−0.003 [0.013]	0.002 [0.012]	−0.011 [0.013]
Constant	0.742*** [0.137]	0.206 [0.202]	0.139 [0.220]	0.381* [0.217]
Observations	558	558	558	554
Number of id	62	62	62	62
Number of instruments	53	61	62	61
AR(2)	0.315	0.161	0.172	0.211
Hansen J test (1)	0.115	0.146	0.162	0.161
Hansen J test (2)	1.000	1.000	1.000	0.300

Notes: The results are estimated by sequential (two-stage) estimation. Robust in brackets.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

low PCI ranking.

As shown in Table 5, Models (1)–(3) are estimated with *GOV*. Models (4)–(6) are estimated with the group of low PCI rankings (*GOV_low*). Models (7)–(9) are estimated with the group of high PCI rankings (*GOV_high*). Observations show that the coefficients on *GOV* are positive, suggesting that public governance has a positive effect on economic growth across provinces (see Columns 1–3). The coefficients on *GOV_low* are significant and negative, suggesting that provinces with low governance levels are less efficient and do not attain economic growth (see Columns 4–6). The coefficients on *GOV_high* are significant and positive, suggesting that provinces with higher governance levels are more efficient and achieve higher economic growth (see Columns 7–9).

To determine whether the effects of PCI's sub-indices on economic growth differ across provinces, the models are estimated with informal charges (*CHARGE*), transparency (*TRANS*), and policy bias (*BIAS*). Specifically, as shown in Table 6, Models (1)–(3) are estimated with informal charges, whereas Models (4)–(6) and (7)–(9) are computed using transparency and policy bias, respectively.

The table shows that the coefficients on *CHARGE* are significant and negative, suggesting that provinces with higher informal charges are less effective and have less economic growth (see Columns 1–3). Similarly, the coefficients on *BIAS* are significant and negative, indicating that provinces with more policy bias are less effective and achieve less economic growth (see Columns 7–9). By contrast, the significant and positive coefficients on *TRANS* imply that provinces with more transparency are more efficient and achieve higher economic growth (see Columns 4–6).

5.4. Economic governance, public spending, private investment, and economic growth

The results reported above show that economic governance, as measured by the PCI, has a significant growth effect. However, it is also of interest to examine how the interactions between public spending, private investment, and economic growth are affected by the governance factors in the PCI. We now turn to determining whether the growth effects of private investment and public spending

Table 4

Effects of combinations of public spending and private investment on provincial economic growth: A disaggregate analysis (Dependent variable: Provincial economic growth, $\Delta \ln(Y)$).

VARIABLES	(1)	(2)	(3)	(4)	(5)
The first stage estimation					
Lag of $\ln(Y)$	−0.133*** [0.024]	−0.140*** [0.029]	−0.142*** [0.024]	−0.147*** [0.023]	−0.130*** [0.023]
$\ln(K)$	0.353* [0.196]	0.471*** [0.176]	0.186** [0.083]	0.162* [0.097]	0.317** [0.158]
$\ln(\text{GINV})$	0.119*** [0.036]	0.075** [0.034]	0.054* [0.032]	0.057* [0.033]	0.066* [0.038]
$\ln(\text{SF})$	0.031* [0.018]	0.030** [0.012]	0.033*** [0.012]	0.037*** [0.013]	0.033** [0.015]
$\ln(\text{FT})$	−0.050** [0.021]	−0.045*** [0.016]	−0.054*** [0.014]	−0.051*** [0.015]	−0.034** [0.014]
$\ln(\text{PUBINV})$	0.492** [0.220]	0.062*** [0.018]	0.024** [0.012]	0.036** [0.016]	0.042** [0.021]
$\ln(\text{PUBINV}) * \ln(K)$	−0.048** [0.024]				
$\ln(\text{CUR})$	−0.028 [0.026]	0.532*** [0.204]			
$\ln(\text{CUR}) * \ln(K)$		−0.061*** [0.023]			
$\ln(\text{ECO})$			0.349** [0.137]		
$\ln(\text{ECO}) * \ln(K)$			−0.037** [0.015]		
$\ln(\text{SCIEN})$				0.596* [0.350]	
$\ln(\text{SCIEN}) * \ln(K)$				−0.065* [0.039]	
$\ln(\text{EDU})$					0.422** [0.212]
$\ln(\text{EDU}) * \ln(K)$					−0.048** [0.023]
Crisis 2008	0.031* [0.019]	0.030* [0.017]	0.059*** [0.013]	0.048*** [0.017]	0.049*** [0.018]
The second stage estimation					
Northern Mountain Region	0.019 [0.012]	0.011 [0.011]	0.014 [0.009]	0.017 [0.011]	0.012 [0.009]
Red River Delta Region	0.031** [0.015]	0.018 [0.014]	0.023 [0.015]	0.020 [0.015]	0.020 [0.014]
Highlands and Central Region	0.005 [0.014]	0.007 [0.013]	0.009 [0.011]	0.007 [0.013]	0.007 [0.011]
Constant	−3.351** [1.671]	−4.132*** [1.555]	−1.296** [0.644]	−1.115 [0.820]	−2.697** [1.371]
Observations	554	554	558	558	558
Number of id	62	62	62	62	62
Number of instruments	62	61	62	61	62
AR(2)	0.399	0.388	0.318	0.315	0.335
Hansen J test (1)	0.164	0.144	0.161	0.142	0.161
Hansen J test (2)	0.998	1.000	1.000	0.997	0.402

Notes: The results are estimated by sequential (two-stage) estimation. Robust in brackets.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Hansen J test (1) is for the first stage estimation.

Hansen J test (2) is for the second stage estimation.

are more pronounced in provinces with good governance.

Using the PCI's sub-indices as above, the interaction terms $\ln \text{PUBINV} * \ln K$, $\ln \text{CUR} * \ln K$, $\ln \text{ECO} * \ln K$, $\ln \text{SCIEN} * \ln K$, and $\ln \text{EDU} * \ln K$ are associated with *CHARGE*, *TRANS*, and *BIAS*, respectively. Table 7, which shows results for Models (1)–(12), indicates that the interaction terms for public spending components and private investment are negative. This suggests a lack of complementarity between public spending and private investment in terms of their effects on economic growth.

Interestingly, when we add three sub-indices of the PCI, we find that only coefficients on the combined terms for public spending components and private investment with transparency (*TRANS*) are significant and positive (see Models (5)–(8)). These results indicate that in provinces with higher levels of transparency, the interaction between public spending and private investment, in terms of their effects on growth, is positive. One significant implication, therefore, is that other dimensions of governance, such as *CHARGE* and *BIAS*, reduce the interactive growth effects emanating from public spending and private investment in Vietnam at the

Table 5Impacts of public governance on economic growth. Dependent variable: Provincial economic growth, $\Delta \ln(Y)$.

VARIABLES	Total PCI ranking			Low PCI ranking			High PCI ranking		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
The first stage estimation									
Lag of $\ln(Y)$	-0.174*** [0.024]	-0.156*** [0.022]	-0.179*** [0.030]	-0.163*** [0.020]	-0.148*** [0.021]	-0.170*** [0.028]	-0.165*** [0.020]	-0.151*** [0.021]	-0.172*** [0.029]
$\ln(K)$	-0.076** [0.036]	-0.067** [0.029]	-0.086** [0.034]	-0.072** [0.031]	-0.065** [0.028]	-0.084*** [0.032]	-0.073** [0.032]	-0.066** [0.029]	-0.085*** [0.033]
$\ln(GINV)$	0.159*** [0.037]	0.163*** [0.038]	0.139*** [0.035]	0.156*** [0.037]	0.160*** [0.039]	0.141*** [0.036]	0.157*** [0.037]	0.161*** [0.039]	0.141*** [0.035]
$\ln(SF)$	0.054*** [0.018]	0.046*** [0.014]	0.061*** [0.018]	0.057*** [0.014]	0.047*** [0.014]	0.066*** [0.018]	0.056*** [0.014]	0.046*** [0.014]	0.065*** [0.018]
$\ln(FT)$	-0.046*** [0.017]	-0.037** [0.016]	-0.057*** [0.020]	-0.045*** [0.015]	-0.037** [0.016]	-0.056*** [0.019]	-0.044*** [0.015]	-0.036** [0.016]	-0.056*** [0.019]
$\ln(PUBINV)$	0.033** [0.014]	0.036*** [0.012]	0.031* [0.017]	0.033** [0.013]	0.035*** [0.012]	0.030* [0.016]	0.033** [0.013]	0.035*** [0.012]	0.030* [0.016]
$\ln(SCIEN)$	0.034** [0.014]	0.038*** [0.014]		0.026* [0.013]	0.033** [0.013]		0.029** [0.014]	0.035*** [0.013]	
$\ln(ECO)$	0.017 [0.025]			0.008 [0.013]			0.010 [0.013]		
$\ln(EDU)$		-0.014 [0.023]			-0.018 [0.023]			-0.016 [0.023]	
$\ln(CUR)$			0.066* [0.038]			0.052 [0.036]			0.055 [0.036]
GOV	0.023** [0.011]	0.016* [0.009]	0.032** [0.014]						
GOV_low				-0.007* [0.004]	-0.006* [0.003]	-0.007* [0.004]			
GOV_high							0.006** [0.003]	0.006** [0.003]	0.007** [0.003]
Crisis 2008	0.055*** [0.017]	0.056*** [0.017]	0.059*** [0.017]	0.058*** [0.016]	0.058*** [0.017]	0.062*** [0.018]	0.057*** [0.016]	0.057*** [0.017]	0.061*** [0.017]
The second stage estimation									
Northern Mountain Region	0.006 [0.010]	0.015 [0.010]	0.004 [0.011]	0.007 [0.010]	0.015 [0.010]	0.001 [0.011]	0.008 [0.010]	0.015 [0.010]	0.002 [0.011]
Red River Delta Region	0.035* [0.019]	0.037* [0.019]	0.041* [0.022]	0.036* [0.020]	0.038** [0.019]	0.039* [0.021]	0.037* [0.020]	0.039** [0.019]	0.041* [0.021]
Highlands and Central Region	-0.004 [0.013]	0.002 [0.012]	-0.011 [0.014]	0.001 [0.013]	0.006 [0.012]	-0.006 [0.014]	0.001 [0.013]	0.006 [0.012]	-0.007 [0.014]
Constant	0.174 [0.229]	0.103 [0.217]	0.134 [0.217]	0.245 [0.202]	0.179 [0.219]	0.285 [0.218]	0.224 [0.201]	0.157 [0.217]	0.252 [0.214]
Observations	558	558	554	558	558	554	558	558	554
Number of id	62	62	62	62	62	62	62	62	62
Number of instruments	63	65	62	63	65	62	63	65	62
AR(2)	0.117	0.132	0.169	0.167	0.170	0.222	0.151	0.155	0.208
Hansen J test (1)	0.161	0.212	0.165	0.161	0.212	0.166	0.161	0.212	0.167
Hansen J test (2)	1.000	1.000	0.787	1.000	1.000	0.787	1.000	1.000	0.785

Notes: The results are estimated by sequential (two-stage) estimation. Robust in brackets

*** p < 0.01, ** p < 0.05, * p < 0.1

Hansen J test (1) is for the first stage estimation

Hansen J test (2) is for the second stage estimation

GOV is proxied by PCI rankings.

Low PCI ranking is for provinces with PCI rankings of 32nd - 63rd.High PCI ranking is for provinces with PCI rankings of first - 31st.

provincial level (see Models (1)–(4) and (9)–(12)). Our results imply that good governance, characterized by attributes including lower informal charges, transparency, and greater accountability, and unbiased policy, is critical for improving the combined growth effect of public spending and private investment across Vietnamese provinces.

5.5. Other variables

Some further observations emerge from [Tables 3–7](#). The estimated results for the lag of income per capita are negative and consistent in all specifications, which suggests that economic growth among provinces in Vietnam might be converging. Second, the central government expenditure has a positive effect on economic growth in Vietnam's provinces in line with [Zhang and Zou \(1998\)](#), implying that the central government, with its advantages in capital investment and knowledge, could efficiently invest in large

Table 6Impacts of public governance on economic growth: An aggregate analysis. Dependent variable: Provincial economic growth, $\Delta \ln(Y)$.

VARIABLES	Informal charges			Transparency			Policy bias		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
The first stage estimation									
Lag of $\ln(Y)$	-0.152*** [0.023]	-0.148*** [0.021]	-0.161*** [0.028]	-0.169*** [0.024]	-0.159*** [0.025]	-0.162*** [0.033]	-0.150*** [0.022]	-0.140*** [0.020]	-0.157*** [0.028]
$\ln(K)$	-0.071** [0.028]	-0.056** [0.027]	-0.085*** [0.025]	-0.078** [0.036]	-0.076** [0.032]	-0.079** [0.033]	-0.068** [0.028]	-0.063*** [0.023]	-0.081*** [0.026]
$\ln(GINV)$	0.114*** [0.042]	0.140*** [0.038]	0.069* [0.040]	0.182*** [0.040]	0.202*** [0.041]	0.181*** [0.035]	0.109*** [0.041]	0.109*** [0.041]	0.087** [0.040]
$\ln(SF)$	0.057*** [0.017]	0.047*** [0.013]	0.069*** [0.017]	0.053*** [0.018]	0.042*** [0.014]	0.050*** [0.017]	0.053*** [0.016]	0.047*** [0.013]	0.068*** [0.018]
$\ln(FT)$	-0.043*** [0.016]	-0.039** [0.016]	-0.057*** [0.020]	-0.051*** [0.018]	-0.045** [0.018]	-0.055*** [0.021]	-0.046*** [0.016]	-0.041** [0.016]	-0.051*** [0.019]
$\ln(PUBINV)$	0.034*** [0.013]	0.035*** [0.012]	0.030* [0.016]	0.034** [0.014]	0.037*** [0.014]	0.040*** [0.016]	0.033** [0.013]	0.035*** [0.012]	0.031** [0.016]
$\ln(SCIEN)$	0.026** [0.012]	0.029** [0.012]		0.026** [0.013]	0.034** [0.015]		0.024** [0.012]	0.029** [0.012]	
$\ln(ECO)$	0.009 [0.022]			0.007 [0.023]			-0.150*** [0.008]		
$\ln(EDU)$		-0.016 [0.022]		-0.020 [0.024]				-0.012 [0.023]	
$\ln(CUR)$			0.061* [0.036]			0.016 [0.039]			0.053 [0.035]
$\ln(GOV)$ _Sub-indices	-0.093*** [0.029]	-0.028* [0.017]	-0.118*** [0.031]	0.122** [0.054]	0.195*** [0.072]	0.161** [0.069]	-0.090*** [0.029]	-0.097*** [0.029]	-0.103*** [0.029]
Crisis 2008	0.036** [0.018]	0.052*** [0.017]	0.045** [0.019]	0.053*** [0.017]	0.050*** [0.017]	0.050*** [0.016]	0.038** [0.018]	0.038** [0.018]	0.040** [0.019]
The second stage estimation									
Northern Mountain Region	-0.007 [0.010]	0.009 [0.009]	-0.018 [0.012]	0.006 [0.010]	0.015 [0.011]	0.011 [0.011]	0.009 [0.009]	0.016* [0.010]	0.003 [0.011]
Red River Delta Region	0.024 [0.019]	0.029 [0.018]	0.024 [0.020]	0.037* [0.020]	0.042** [0.021]	0.043** [0.020]	0.036** [0.018]	0.037** [0.018]	0.039* [0.020]
Highlands and Central Region	-0.013 [0.013]	-0.001 [0.011]	-0.024* [0.014]	-0.006 [0.013]	-0.001 [0.013]	-0.007 [0.014]	0.003 [0.013]	0.008 [0.013]	-0.005 [0.014]
Constant	0.682*** [0.261]	0.310 [0.209]	0.967*** [0.267]	-0.074 [0.268]	-0.325 [0.274]	-0.226 [0.254]	0.663*** [0.254]	0.649*** [0.249]	0.754*** [0.256]
Observations	558	558	554	558	558	554	558	558	554
Number of id	62	62	62	62	65	62	62	62	62
Number of instruments	63	65	62	63	62	62	63	65	62
AR(2)	0.117	0.132	0.227	0.147	0.130	0.200	0.214	0.214	0.296
Hansen J test (1)	0.161	0.212	0.165	0.161	0.212	0.165	0.161	0.212	0.161
Hansen J test (2)	1.000	1.000	0.989	1.000	1.000	0.706	1.000	1.000	0.582

Notes: The results are estimated by sequential (two-stage) estimation. Robust in brackets.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Hansen J test (1) is for the first stage estimation.

Hansen J test (2) is for the second stage estimation.

Sub-indices include informal charges, transparency and policy bias, respectively.

Models 1-3 are estimated for informal charges.

Models 4-6 are estimated for transparency.

Models 7-9 are estimated for policy bias.

public projects, which are supportive of economic growth across provinces. Among Vietnam's regions, the time-invariant factors of the Red River Delta region have positive effects on economic growth in this region. The 2008 financial crisis variable has a positive effect on provincial economic growth, implying that, after the 2008 financial crisis, Vietnamese provincial governments designed proper strategies to overcome the challenges and difficulties due to the crisis and restore and promote economic growth across provinces.

6. Conclusions

The main objective of this paper is to contribute to the existing literature on the relationships among public spending, economic governance, private investment, and economic growth. The empirical model is the growth function of the province constructed on the assumption that the central government can intervene in provincial growth through public investment policy, fiscal decentralization, and transfers. Local governments can stimulate provincial growth by enhancing self-financing, providing local public goods/services,

Table 7

The role of public governance in improving the effect of private investment on economic growth: An aggregate analysis Dependent variable: Provincial economic growth, $\Delta \ln(Y)$.

	Informal charges			Transparency			Policy bias					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
The first stage estimation												
Lag of ln (Y)	-0.164*** [0.027]	-0.142*** [0.021]	-0.173*** [0.022]	-0.182*** [0.023]	-0.170*** [0.030]	-0.164*** [0.033]	-0.154*** [0.025]	-0.166*** [0.030]	-0.163*** [0.030]	-0.141*** [0.022]	-0.124*** [0.023]	-0.148*** [0.020]
Ln(K)	0.315*** [0.116]	0.412*** [0.130]	0.109* [0.060]	0.162** [0.077]	0.409** [0.174]	0.811*** [0.216]	0.085* [0.049]	0.135** [0.064]	0.357** [0.162]	0.553*** [0.166]	0.096* [0.054]	0.155* [0.081]
Ln(GINV)	0.083* [0.045]	-0.007 [0.052]	0.073* [0.041]	0.110** [0.043]	0.088* [0.053]	0.127** [0.063]	0.047 [0.037]	0.114*** [0.040]	0.080** [0.036]	0.007 [0.039]	0.012 [0.031]	0.037 [0.032]
Ln(SF)	0.067*** [0.024]	0.029*** [0.013]	0.055** [0.022]	0.081*** [0.018]	0.034** [0.014]	0.042*** [0.015]	0.052** [0.025]	0.039*** [0.012]	0.056** [0.024]	0.024* [0.013]	0.024* [0.013]	0.030*** [0.011]
Ln(FT)	-0.040** [0.021]	-0.058*** [0.017]	-0.049*** [0.017]	-0.039** [0.020]	-0.062*** [0.020]	-0.040** [0.020]	-0.044** [0.021]	-0.051*** [0.018]	-0.041** [0.019]	-0.057*** [0.017]	-0.047*** [0.014]	-0.058*** [0.018]
Ln(PUBINV)	0.412*** [0.137]	0.017* [0.010]	0.032* [0.017]	0.041*** [0.014]	0.468** [0.191]	0.025** [0.012]	0.037** [0.016]	0.029** [0.012]	0.433** [0.182]	0.067*** [0.032]	0.017* [0.010]	0.028** [0.011]
Ln(CUR)	0.017 [0.032]	0.501*** [0.158]			0.011 [0.028]	0.931*** [0.258]			-0.007 [0.038]	0.590*** [0.191]		
Ln(ECO)			0.240** [0.106]				0.191** [0.090]				0.211*** [0.075]	0.570* [0.308]
Ln(SCIEN)				0.626** [0.289]				0.525** [0.250]				
Ln(K)* Ln(PUBINV)	-0.039** [0.017]				-0.091** [0.036]	-0.169*** [0.036]	-0.000 [0.000]		-0.037* [0.019]	0.004 [0.015]	-0.001* [0.000]	
Ln(K)* Ln(CUR)		-0.052*** [0.017]										
Ln(K)* Ln(ECO)			-0.000 [0.000]									
Ln(K)* Ln(SCIEN)				-0.067** [0.032]				-0.128*** [0.036]				-0.050* [0.030]
Sub-indices	0.105 [0.348]	-0.006 [0.061]	0.617** [0.269]	-0.057 [0.055]	-1.664** [0.811]	-2.523*** [0.870]	0.433* [0.226]		0.239* [0.137]	0.210 [0.142]	0.373** [0.174]	0.142 [0.093]
Ln(K)* Ln(PUBINV)*Sub-indices	-0.001 [0.005]				0.023** [0.012]				-0.004** [0.002]			
Ln(K)* Ln(CUR)*Sub-indices		-0.072*** [0.027]				0.037*** [0.013]				-0.003* [0.002]		
Ln(K)* Ln(ECO)*Sub-indices			-0.011** [0.005]				-0.009* [0.005]				-0.008** [0.003]	-0.007* [0.004]
Ln(K)* Ln(SCIEN) *Sub-indices				-0.049*** [0.019]				0.040*** [0.016]				0.065*** [0.018]
Crisis 2008	0.044** [0.020]	0.036* [0.021]	0.057*** [0.019]	0.018 [0.025]	0.039** [0.018]	-0.002 [0.021]	0.057*** [0.019]	0.040** [0.017]	0.046** [0.020]	0.054*** [0.019]	0.092*** [0.024]	0.065*** [0.018]
The second stage estimation												
Northern Mountain Region	0.031*** [0.011]	-0.372 [0.227]	0.020*** [0.008]	-0.035 [0.103]	0.007 [0.008]	0.010 [0.009]	0.017* [0.009]	0.008 [0.009]	0.032*** [0.011]	0.010 [0.008]	0.008 [0.007]	0.009 [0.009]
Red River Delta Region	0.020 [0.016]	0.110 [0.219]	0.015 [0.017]	0.157 [0.140]	0.020 [0.012]	0.021* [0.012]	0.011 [0.017]	0.023 [0.015]	0.014 [0.016]	0.014 [0.013]	0.014 [0.014]	0.018 [0.015]

(continued on next page)

Table 7 (continued)

	Informal charges			Transparency			Policy bias					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Highlands and Central Region	0.002	-0.403	0.009	-0.161	-0.001	0.005	0.000	0.001	0.004	0.011	0.004	0.007
	[0.015]	[0.251]	[0.013]	[0.117]	[0.011]	[0.012]	[0.014]	[0.012]	[0.015]	[0.010]	[0.011]	[0.012]
Constant	-2.981**	6.530	-1.934*	0.808	-0.442	-2.878	-1.235	0.336	-3.340**	-4.602***	-1.010	-1.071
	[1.209]	[3.973]	[1.133]	[0.963]	[1.273]	[2.247]	[0.779]	[0.747]	[1.544]	[1.544]	[0.742]	[0.784]
Observations	554	554	554	558	554	554	554	558	554	554	554	558
Number of id	62	62	62	62	62	62	62	62	62	62	62	62
Number of instruments	63	63	63	63	67	63	61	63	67	63	63	63
AR(2)	0.402	0.347	0.428	0.323	0.372	0.739	0.313	0.293	0.465	0.406	0.474	0.168
Hansen J test (1)	0.139	0.139	0.139	0.139	0.240	0.139	0.101	0.139	0.240	0.139	0.139	0.139
Hansen J test (2)	1.000	0.313	0.157	0.916	1.000	1.000	0.123	1.000	0.847	0.321	1.000	1.000

Notes: The results are estimated by sequential (two-stage) estimation. Robust in brackets.

*** p < 0.01, ** p < 0.05, * p < 0.1.

Hansen J test (1) is for the first stage estimation.

Hansen J test (2) is for the second stage estimation.

Sub-indices include informal charges, transparency and policy bias, respectively.

Models 1-4 are estimated for informal charges.

Models 5-8 are estimated for transparency.

Models 9-12 are estimated for policy bias.

and improving local governance. The following findings emerge from our empirical study, based on data from sixty-two provinces in Vietnam over the period 2006–2015.

First, the disaggregated analysis shows that public expenditure components have inconsistent effects on economic growth. The effect of expenditure on infrastructure and technology is significantly positive whereas that of expenditure on education and economic services is negative and current expenditure has an insignificant effect. In addition, the interaction terms for public spending and private investment are negative with regard to their association with economic growth. These findings, combined with others reported in the paper, suggest that the marginal benefit for economic growth from increased Vietnamese provincial government expenditures may be limited in the absence of improvement in the quality of governance, which has been shown to enhance interaction between government spending and private sector investment. This merits greater consideration, given the increase in public expenditure at the provincial level together with the movement toward improved decentralization and public governance in Vietnam.

Second, the PCI ranking, as a proxy for provincial economic governance, has been confirmed as having a significant effect on economic growth in Vietnam's provinces. To understand how governance conditions the effects of public spending, the study uses PCI sub-indices, namely informal charges, transparency, and policy bias, which reflect attributes of good governance. We find that the growth effects of public expenditure and private investment are all strengthened in the provinces with greater transparency. In other words, informal charges and policy bias do not generate growth effects from public expenditure and private investment. This indicates that less corruption (or informal charges), transparency and greater accountability, and unbiased policy are associated with growth in the private sector by enhancing the positive linkages between government and private sector activity. In recent years, PCI scores between the provinces have tended toward convergence, driven largely by major improvements in the provinces with the poorest performance (in fact, these provinces, located mainly in the northern mountainous and central regions, have commonly emulated the best practices in enhancing the business environment). Provinces with a PCI ranking in the top half have faced greater challenges in innovating new governance measures to promote economic activity (Malesky et al., 2017).

The results reported in this paper are subject to the usual caveats related to data quality and availability and also interpretations of econometric modeling. Therefore, it will be interesting to see whether similar findings are reached with different panel-data sets and with alternative econometric techniques. Overall, this study sheds light on the nature and significance of the role of economic governance in influencing the effectiveness of growth outcomes associated with public expenditure and private investment. Although public expenditure and private sector investment spending make important contributions to economic growth, the quality of economic governance is critical in reinforcing positive interactions between the expenditure categories in a manner that promotes economic growth. Under fiscal decentralization, a great deal of public spending on education, health, and infrastructure is determined by block grants from the central level, whereas economic governance is measured by policy levers controlled by local leaders. Therefore, the potential for enhanced and sustained economic growth and development in Vietnamese provinces depends heavily on their capacity to pursue and achieve even higher levels of governance.

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Declaration of Competing Interest

The authors report no declarations of interest.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.ecosys.2020.100780>.

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